

STUDENT MANAGEMENT SYSTEM

Interface Design & Figma UI Prototype

Role-based interface for Admin, Teacher and Student

PRESENTER

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SECTION

Interface Design

SYSTEM

C++ Console + MySQL

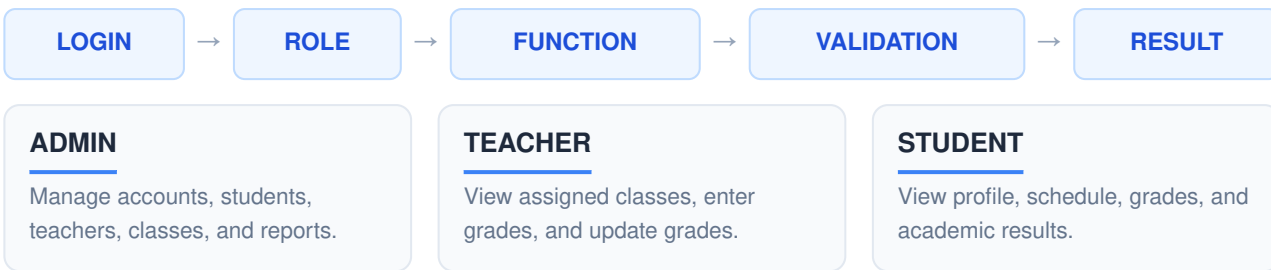
PROTOTYPE

Figma

02 / 08

1. Interface Design Overview

Current implementation: C++ Console | Visual prototype: Figma



Simple menus • Clear permissions • Immediate feedback

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2. Login and Role Authentication

- **Valid credentials:** Identify role → open the correct menu for workspace access.
- **Invalid credentials:** Return the user immediately to Login Screen with error feedback.
- **Strict permissions:** Each role only accesses its permitted functions and resources.

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3. Admin Interface and Student Management

Admin uses a structured hierarchical menu: main functions → detailed operations.

Module	Primary Functions	Key Operations
Admin Dashboard	System overview and quick access	View statistics (1,284 Students, 86 Teachers, 42 Classes, 18 Subjects)
Manage Students	Search, add, update and review records	Filter by ID/Name, add new student profiles, edit details, set status

05 / 08 4. Teacher Grade Entry and Validation

Teachers select Class, Subject, and Semester to load and manage student grade lists.

Validation Rule	Condition / Criteria	System Feedback
Required Fields	Fields cannot be empty	Error: Field Required
Score Range	Score must be between 0.0 and 10.0	Invalid: Must be 0 - 10
Data Format	Numeric values only	Success: Saved Successfully

06 / 08 5. Student Academic Results

- Students view only their personal profile, schedule, grades, and overall academic performance (e.g., Cumulative GPA: 8.65 - Excellent).
- **Security & Integrity:** Zero permissions to modify grades or access another student's confidential information.

07 / 08 6. System Integration and Docker

Request and Data Flow Architecture:

UI (Console / Figma) → CONTROLLER (Navigation) → SERVICE (Business rules) → REPOSITORY (Data access) → MYSQL (Persistent data)

- **System Integration:** Connects user requirements, UI, business logic, modules, and database into one consistent system.
- **Docker Containerization:** Standardizes the environment and dependencies. If no Docker files exist yet, this remains an implementation plan.
- **Workflow:** Branch → Pull Request → Review → develop. (Docker complements MySQL without replacing it).

The interface is designed around role segregation, robust validation, and immediate user feedback.

- **ROLE-BASED** Admin, Teacher, and Student have clear, isolated permissions.
- **VALIDATED** All inputs are strictly checked prior to database persistence.
- **EXTENDABLE** Figma visualizes a future graphical UI built on top of the C++ core.

Interactive Figma Prototype Link

[Open Figma Prototype](#)

THANK YOU